

On moral responsibility

A metaphysical examination

Alexander Towell
lex@metafunctor.com

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1 Introduction

People throughout history have believed that they belong to a special categorical class called *persons*. What supposedly makes members of this class special are their status as *moral agents*. But why do they warrant this distinction? Likely, it is based on criteria such as being able to make choices, form intentions, and perform actions. However, if the nature of reality is at odds with such criteria, our moral agency could be reduced to a pleasant fiction.

Metaphysics While challenges to the proposition that persons are moral agents have come from many different disciplines, few have posed a greater challenge to it than metaphysics. Metaphysics is a discipline in philosophy with a very long history. The etymology of *metaphysics* derives from the Greek words *metá* (after) and *physiká* (physics), and it simply refers to Aristotle's work that contiguously followed after his chapter on physics. However, over time the meaning of *metá* has been reinterpreted to mean something more substantial, like *beyond*. Thus, metaphysics is the study of that which is beyond physics. And arguably it is the most fundamental of all disciplines for it studies the categorical structure of reality itself.

There is some disagreement among meta-physicians about this categorical structure. Disagreements are not just over things like the kinds of categories that ought to be recognized, but they are also over things like whether reality even has a categorical structure. That is, do categories represent absolute ontological differences or do they represent the useful ways in which humans organize their experiences? Does reality have an inherent categorical structure (which may

someday be discovered) or do minds invent conventional categories to order their experiences?

Moral agency To say that we are moral agents is to say that we can be held morally responsible for our actions, which can run the gamut from *good* to *bad*. For example, people who are around a village during a volcanic eruption may demonstrate good moral conduct if they rescue villagers from the destructive power of the volcanic eruption. Alternatively, they may demonstrate bad moral conduct if they do nothing to help the villagers. The volcano eruption is causally responsible for the destruction to the village, but it is not morally responsible.

Under what circumstances are things morally responsible? A theory of moral responsibility might touch upon, at the very least, the following subjects:

1. The subject of morality itself.
2. The criteria for being a moral agent.
3. The circumstances under which moral responsibility can be ascribed.

In due course, after expanding upon each of these points, one might arrive at a satisfactory explanation that accounts for moral responsibility. However, the discipline of metaphysics has not only undermined the notion that persons are morally responsible, but also the notion that reality can even support the possibility of moral responsibility.

2 Morality

“Morality is simply the attitude we adopt towards people whom we personally dislike.” –Oscar Wilde

The etymology of *morality* derives from the Latin word *moralitas* (manner, character, proper behavior), and it can be used a) descriptively to refer to a code of conduct promoted by some group, or b) normatively to refer to a code of conduct that, given specified conditions, would be promoted by any moral agent. In what follows, our primary concern will be of the second normative type where the nature of morality can be explored in greater isolation from cultural norms.

What is the basis for morality? From a behavioral point of view, morality is arguably the study of right and wrong conduct. However, what is the basis of right and wrong? One possibility is that its basis is authoritative prescription, in which case morality is a dictum of *might makes right*—something is morally good or bad because an authority says so with no further explanation required. A forceful criticism of this type of moral theory is revealed by asking: is an action good because God commands it, or does God command it because it is good? If it is good simply because God commands it, then morality is nothing more than the arbitrary preference of God. But if God commands it because it is good, then morality has a basis independent of God.

Many would contend the basis is essentially rational; persons use reason and experience to inform their conduct, e.g., how can a group best realize common interests? In which case, a sense of right and wrong is, for the most part, the habituation of conduct that promotes cultural dominance or health, i.e., the principle of natural selection applied in the context of a social Darwinism.

While there are perhaps many other moral theories—some of which may not support the premise of moral responsibility—each must fall within one of two general categories: the conventional, and the intrinsic. On the one hand, if morality is conventional, then persons are only morally responsible by convention. On the other hand, if morality is categorically intrinsic, then arguably moral agents have an inherent duty to conform to it. But what does it mean to say that morality is categorically intrinsic?

Realism

There are many things—particulars—which are red. For instance, there are red cars, red apples, and red signs. But *what* makes them red? The philosophical discipline of realism submits that all red particulars are red in virtue of the existence of a Universal—a single abstract entity that is a part of all red particulars. More precisely, Universals are repeatable entities that can be exemplified simultaneously by different particulars, where entities are things like *properties*, like *red*, *relations*, like *behind*, and *kinds*, like *dog*.

Realists argue that attribute agreement is due to relationships between entities and Universals. Furthermore, they contend that the subject-predicate grammar of sentences reinforces this view, e.g., in the sentence “My apple is red,” the predicate “is red” seems to be picking out something (a noun)—call it *redness*—and linking it to the subject, “my apple.” Let us examine how a realist might account for following two sentences:

1. My apple is red.
2. My friend’s apple is red.

In each sentence, there are arguably two Universals being exemplified. First, there is the Universal property, *redness*, that is being exemplified by the two particulars, *my apple* and *my friend’s apple*. Second, there is the Universal kind, *apple-ness*, that is also being exemplified by *my apple* and *my friend’s apple*. The *redness* being exemplified by the entity, *my apple*, is numerically identical with the *redness* being exemplified by the entity, *my friend’s apple*. Likewise, the *apple-ness* that is being exemplified by *my apple* is numerically identical with the *appleness* that is being exemplified by *my friend’s apple*.

In the sentence, “This apple is red,” a metaphysical realist maintains that this denotes “*This apple* exemplifies *redness*,” where *redness* is a Universal that inheres in *this apple*. More generally, a realist claims that any subject-predicate sentence of the form “*a* is *F*” can be paraphrased as “*a* exemplifies *F-ness*” where *a* is an entity and *F-ness* is a Universal. In the same way, a realist could maintain that *morality*, like *color*, is a Universal. In the same way that an apple

can exemplify a color like *redness*, a person can exemplify a *virtue* (a moral excellence) like *honesty*, e.g., “Bob exemplifies honesty.” However, it turns out that the existence of Universals is one of the most important disputes in the discipline of metaphysics.

Paradox If “*a* is *F*” denotes “*a* exemplifies *F*-ness,” then we can construct a paradox. Let *F* mean “does not exemplify itself” and so *F*-ness could be the Universal property, *non-selfexemplification*. Thus, for some *a*, the proposition “*a* exemplifies *non-selfexemplification*,” should be either **true** or **false**. But the existence of this property leads immediately to a paradox, for if a thing does exemplify itself, then it does not exemplify itself and if a thing does not exemplify itself, then it does exemplify itself. It is self-contradictory.

On the one hand suppose that an entity, *a*, does exemplify itself. Then, since it is a property an entity exemplifies when it does not exemplify itself, *a* does not exemplify itself. On the other hand, suppose that *a* does not exemplify itself. Then, it does exemplify itself. It is similar to the sentence, “This sentence is a lie.” If it is **true**, then it is **false**; and if it is **false**, then it is **true**. It is self-contradictory.

To avoid this paradox, restrictions on what constitutes a Universal must be imposed, such as denying that *non-selfexemplification* is an actual Universal. However, as a result of this, the realist cannot take it for granted that “*a* is *F*” means that *F* must be selecting a Universal, *F*-ness. This is not necessarily a problem for morality, but the realist cannot simply claim, without qualification, that if a sentence can take on a subject-predicate form that necessarily means that the existence of Universals can be assumed to account for things like attribute-agreement. For instance, “*Socrates is honest*” does not necessarily mean that the adjective, *honest*, is pick out out a Universal entity, *honesty*.

Bradley’s regress: The central challenge to realism A more fundamental challenge to realism emerges when we examine the nature of the exemplification relation itself. This problem, first articulated by F.H. Bradley in his 1893 work *Appearance and Reality* [2], strikes at the heart of how Universals are supposed to relate to particulars. Bradley argued that if relations are themselves Universals, then an infinite regress ensues that may undermine the entire realist project.

Consider the realist’s analysis of “Socrates is courageous” as “Socrates exemplifies courage,” where courage is a Universal property and Socrates is a particular. This analysis invokes three entities: Socrates, the Universal courage, and the exemplification relation that connects them. But if exemplification is itself a Universal relation, then we must explain how this relation relates to its relata—how exemplification itself connects to both Socrates and courage.

It appears we need a further relation, exemplification₂, to relate exemplification₁ to Socrates and courage. But then exemplification₂ requires exemplification₃ to relate it to its relata, and so on ad infinitum. For any subject-predicate sentence “*a* is *F*,” the realist analysis generates an infinite hierarchy:

1. *a* exemplifies₁ *F*-ness
2. The complex “*a* exemplifies₁ *F*-ness” exemplifies₂ exemplification

3. The complex “[a exemplifies₁ F -ness] exemplifies₂ exemplification” exemplifies₃ exemplification
4. And so on, indefinitely

The critical question is whether this infinite regress is vicious—that is, whether it undermines the realist’s explanatory project. An infinite regress is generally considered vicious when each step in the series depends on a subsequent step for its explanation or grounding, such that the explanation can never be completed.

Realist responses to Bradley’s regress Contemporary realists have developed several sophisticated strategies to address this challenge, and the success or failure of these responses is central to assessing realism about moral properties.

The non-relational tie response. The most influential realist response, defended prominently by Armstrong [1], denies that exemplification is a genuine Universal. Instead, exemplification is a “non-relational tie” or “fundamental nexus”—a basic feature of reality that requires no further analysis. On this view, that particulars instantiate Universals is a brute metaphysical fact. The basic furniture of reality includes not just particulars and Universals, but also the instantiation relation that holds between them. Just as we must accept some primitives in any explanatory framework (e.g., physical laws in science, logical connectives in logic), the realist claims we should accept instantiation as primitive.

Critics object that this response abandons the realist’s explanatory ambitions. If instantiation is primitive and inexplicable, then the realist has not really explained attribute agreement—they have merely labeled it. The nominalist might retort: if you are willing to accept primitive facts about instantiation, why not accept primitive facts about resemblance instead, thereby avoiding commitment to Universals altogether?

The benign regress response. An alternative strategy, explored by contemporary metaphysicians like Maurin [13], accepts the infinite regress but denies its viciousness. The argument proceeds as follows: once we establish the ground-level fact that a exemplifies F -ness, all higher-order facts in the infinite series follow automatically and require no additional metaphysical grounding. The truth of “[a exemplifies₁ F -ness] exemplifies₂ exemplification” is not an independent fact requiring explanation; it is simply a consequence of the first-order fact plus the general principle that when a particular instantiates a Universal, that instantiation itself instantiates the Universal of exemplification.

This response faces its own difficulties. It must explain why the regress, though infinite, is not vicious. One criterion often proposed is that a regress is benign if each level is explanatorily posterior to the previous level—that is, the higher-order facts are explained by the lower-order facts, not vice versa. But this seems to contradict Bradley’s original worry: if the higher-order exemplification facts are needed to ground the claim that a exemplifies F -ness, then the explanation runs upward, not downward, making the regress vicious.

Criteria for vicious regress Following the contemporary literature [13], an infinite regress is generally considered vicious when it exhibits one or more of the following features:

1. *Explanatory circularity*: Each level in the regress requires the subsequent level for its explanation, such that no level provides a foundation (the explanation “chases its tail” infinitely)
2. *Ontological dependence*: Each fact or entity at level n depends for its existence on a distinct fact or entity at level $n + 1$, creating an infinite chain of dependencies
3. *Epistemic inaccessibility*: To know that p_1 is true, we must first know that p_2 is true, but to know that p_2 is true we must first know that p_3 is true, ad infinitum, making knowledge impossible

A regress is benign if the infinite series exists but does not exhibit these problematic features—for instance, if all higher-level facts follow automatically from a ground-level fact without requiring independent verification or grounding.

Implications for moral properties The Bradley regress poses special difficulties for moral realism specifically. If the regress undermines realism about Universals generally, then moral properties like virtue, honesty, courage, and responsibility cannot be grounded as objective features of reality. The realist moral philosopher who claims that “Torture is wrong” expresses the objective fact that torture exemplifies the Universal property of moral wrongness faces all the problems discussed above.

Moreover, moral properties present an additional challenge: they seem to require not just exemplification but also a normative force—a reason-giving or action-guiding dimension. Even if we could solve Bradley’s regress for descriptive properties like redness or roundness, it remains unclear whether this solution extends to normative properties. The statement “This action exemplifies wrongness” must not only pick out an objective property but must also somehow explain why this property gives us reason to avoid the action.

If Bradley’s regress is vicious, moral realism collapses into nominalism or conventionalism about moral properties. On such views, moral categories would be useful classifications without ontological significance—more akin to legal categories (which vary by jurisdiction and convention) than to natural kinds. The implications for moral responsibility are profound: if “moral agent” is a conventional designation rather than a recognition of metaphysical fact, then ascriptions of moral responsibility become matters of social practice rather than objective truth. We might still hold people responsible, but this practice would lack the metaphysical foundation many believe it requires.

Nominalism

There is competing philosophical doctrine to realism, called nominalism, which disclaims the existence of Universals. Nominalists believe that realism needlessly introduces the strange notion of Universals, exhibits incoherency, and is burdened by a busy ontology. In its place they propose a supposedly simpler but sufficient account for the apparent categorical structure of reality.

Nominalists submit that only actual particulars have independent existence. After all, how could a repeatable entity, like *redness*, simultaneously inhere in multiple entities, like an apple and a firetruck, such that the *redness* in the apple is numerically identical to the *redness* in the firetruck? The Realist is introducing a strange idea that entities with independent existence have non-local, unbounded physical presence. So, realists claim that the sentence, “*a* is *F*” denotes “*a* exemplifies *F-ness*” where *F-ness* is a Universal. In response, nominalists counter that “*a* is *F*” denotes “*a* is a member of the set of *F things*.” For instance, “*Socrates* is courageous” denotes “*Socrates* is a member of the set of *courageous things*.” Nominalists introduce *sets* (which are rigorously defined mathematical entities) to do the work of Universals. And the primary pay-off to this is that they avoid having to invoke the strangeness of Universals.

Set An unordered collection of entities considered as a whole. The identity conditions for sets is such that, if *a* and *b* are sets, they are numerically identical iff for each entity in *a* that entity is in *b*. Using mathematical language, $a \equiv b$.

Abstract reference For simple subject-predicate sentences, the nominalist’s account as already given would seem sufficient. However, how might they interpret a sentence including abstract parts, like “Honesty is a virtue?” The realist would contend that this can be understood to mean, “*Honesty* exemplifies being a *virtue*,” *honesty* and *virtue* picking out abstract Universals. But what might nominalists say? One possible interpretation might be, “The set of *honest acts* is a subset of the set of *virtued acts*.” While this gives nominalists a little trouble, their response might seem sufficient enough. However, their problems do not end here.

Set equivalency How might a nominalist deal with the supposition that all the things that have hearts also have kidneys and vice versa? In other words, the set of *things that have hearts* and set of *things that have kidneys* have the same members. But according to the identity conditions for sets, that means that the set of *things that have hearts* is identical to the set of *things that have kidneys*. Consequently, *having a heart* is the same thing as *having a kidney* (the set of *things that have hearts* $\equiv$ the set of *things that have kidneys*), which is plainly absurd.

What is the nominalist’s response to this? Generally, they invoke the terminology of *possible* entities. Instead of “*a* is *F*” denoting “*a* is a member of the set of *F-things*,” it denotes “*a* is a member of the set of *possible F-things*.” The set of *possible things that have hearts* and set of *possible things that have kidneys* do not have the same members, therefore they are not equivalent. Unfortunately,

part of the justification for nominalism was that it has a simpler ontology than realism, but the introduction of *possible worlds* somewhat undermines this effort.

Conventionalism Even if nominalism did offer a satisfactory account of attribute agreement and subject-predicate discourse, it is none-the-less nominative. Accordingly, there is no entity, *honesty*, there are only sets of resembling things. The nominalist posits that the sentence, “My friend is honest” simply means that *my friend* resembles other persons that are called honest. This does not appear to give ontological significance to *honesty*. Thus, we come back to where we started: if right and wrong is conventional, then is being a moral agent ontologically significant?

From metaphysical foundations to agency Having examined the metaphysical status of moral properties themselves, we must now turn to the agents who supposedly bear moral responsibility. Even if we were to resolve the debate between realists and nominalists about moral properties, a separate but equally fundamental question remains: what are the criteria for being a moral agent? The metaphysical challenges we have explored regarding Universals and conventions now resurface in our examination of personhood and agency. If moral properties lack objective grounding, and if the agents themselves are metaphysically problematic, then the entire edifice of moral responsibility becomes questionable.

Criteria for moral agency

“I know only that what is moral is what you feel good after and what is immoral is what you feel bad after.” –Ernest Hemmingway

What is the criteria for being a moral agent? By definition, a moral agent is a being who can make moral judgments. Thus, at a minimum, it would seem a moral agent should have the capacity to make rational decisions—that is, reason about the effects their decisions have on reality.

Core requirements for agency The traditional philosophical account of moral agency requires several interconnected capacities. First, the agent must possess *rationality*—the ability to form beliefs based on evidence, draw logical inferences, and understand causal relationships. Second, the agent requires *intentionality*—the capacity to form intentions and direct actions toward specific goals. Third, the agent needs *reflective awareness*—the ability to consider one’s own mental states and evaluate one’s actions against moral principles.

Consider a concrete example: a physician deciding whether to administer an experimental treatment to a terminally ill patient. The physician must rationally evaluate the evidence for the treatment’s efficacy, form an intention based on the patient’s best interests, and reflect on the moral implications of the decision. Without any one of these capacities, genuine moral agency becomes questionable.

Degrees of moral responsibility Rather than treating moral agency as a binary property, we might consider it as existing along a spectrum. Young

children, for instance, gradually develop moral agency as their cognitive capacities mature. They begin by understanding simple cause-and-effect relationships (“hitting hurts others”), progress to recognizing intentions (“I didn’t mean to break it”), and eventually develop the capacity for moral reflection (“Was my action fair?”).

This graduated view raises important questions about the boundaries of moral responsibility. If a person’s rational capacities are temporarily impaired—through intoxication, extreme emotional distress, or sleep deprivation—to what extent are they morally responsible for their actions? The law recognizes such gradations through concepts like diminished capacity and mitigating circumstances, but the philosophical foundations remain contentious.

The knowledge condition Beyond basic cognitive capacities, moral agency seems to require adequate knowledge of relevant circumstances. An agent cannot be held fully responsible for consequences they could not reasonably have foreseen. This epistemic requirement introduces complications: what constitutes “reasonable” foresight? Should moral agents be held responsible for their ignorance itself?

Consider the historical example of physicians who prescribed mercury as medicine before its toxic effects were understood. Were they morally responsible for the resulting harm? The answer depends partly on whether the relevant knowledge was available and whether they had an obligation to seek it. This suggests that moral agency includes not just the capacity to make informed decisions but also the responsibility to become adequately informed.

So, how do persons measure up to these criteria? At the very least, they seem to have the capacity to reason about a limited domain of reality, and so are perhaps morally responsible within the limits circumscribed by that domain. But putting all of that aside, one can instead ask, “Do moral agents exist at all?” That is, are there subjects *making* decisions?

Persistence

To consider this question, let us analyze Chisholm’s essay, *Identity through Time* [3]. In this essay, he distinguishes two ways in which entities persist.

Intactly persisting entities In the first way, entities persist in a numerically identical sense such that an entity, a , at time t_1 is the same as entity, b , at time t_2 iff the set of parts for entity a at time t_1 is equivalent to the set of parts for entity b at time t_2 . Therefore, if $a \equiv b$, said entity persisted, at a minimum, $|t_2 - t_1|$ units of time.

Non-intactly persisting entities If an entity changes even a single part, it is not strictly the same. Therefore, since physical entities are in a constant state of flux, when one says physical entity a at time t_1 is the same as physical entity b at time t_2 (where $t_2 \neq t_1$), we do not mean that entities a and b are numerically identical. Rather, we mean to say that they are conventionally the same.

Ship of Theseus To clarify, let us consider the thought experiment of the *Ship of Theseus*, which raises the question of whether a ship which has had all

its parts replaced retains the same identity. If the ship at t_1 has different parts than the ship at t_2 then they are not numerically identical; so, if the ships are identical, a non-intactly persisting sense of identity is being used. Chisholm contends that the question is not *is it the same ship* but *does it constitute the same ship?* And if it constitutes the same ship, that means it satisfies the Theory of Persistence.

Theory of Persistence A composite entity persists if conditions 1 and 2 are satisfied:

1. The precedent composite entity evolved from a antecedent composite entity. Note: Evolved denotes that after each successive change, the antecedent composite must have at least one part in common with the precedent composite.
2. The composite entity must satisfy additional quantifiable criteria.

In practice, how might this theory provide a solution for the *Ship of Theseus* dilemma? First, we must determine whether it satisfies, from initial state to final state, the *evolution* criterion. And second, it must satisfy additional criteria, e.g., *has the same sailing schedule*. For example, if the evolution of the ship undergoes fission then the *evolution* criterion by itself cannot determine which ship constitutes the *Ship of Theseus*. However, if only one of the ships after the fission has the same sailing schedule as the ship before the fission, then according to the *has the same sailing schedule* convention, that ship constitutes the *Ship of Theseus*.

Persons

It is not even clear how *persons* should be defined, but providing an unambiguous definition for it would help to resolve many controversial. For instance, to resolve the dispute over abortion, one could categorically determine when a bundle of cells constitutes person-hood. There appears to be a consensus that motile sperm cells and ova are not persons, but that newborn babies are. What was the critical turning point in which it from just being a bundle of cells to being a person who has a bundle of cells? Expending considerable time and effort could be avoided if a definitive categorical account of this problem was given, but like with the pile of sand, such a definition seems problematically conventional.

Chisholm on persons In light of the *Theory of Persistence*, what can be said about persons? As with the *Ship of Theseus*, are person's identities also conventional? Or are they like sub-atomic particles which intactly persist? In *Which Physical Thing Am I* [4], Chisholm contends that *persons* are not conventional. In support of this, he asks you the reader to consider a hypothetical situation in which you are about to undergo an operation. The doctor presents

you with two options: you can either choose the expensive option where you are subjected to total anesthesia during the operation, or you can choose the cheaper option. The cheaper option consists of the doctor giving you a pill before the operation to induce complete amnesia so that during the operation there is no memory of you, and giving you a pill after the operation so that everything that transpired during the operation is forgotten.

The question is, would it be reasonable for you to choose the cheap option? Chisholm contends that it would not be reasonable because you would believe that you would be the person enduring the operation. Furthermore, you would not be persuaded that this is not the case by any convention that denies that the person on the table is you. In other words, you would not consider your identity a matter of convention.

Critical response to Chisholm How might Chisholm’s arguments be disputed? In the situation he presented, perhaps you would believe that you and the person enduring the operation in the cheaper option is the same person. Furthermore, perhaps no convention could persuade you from this conviction. However, your conviction could simply be based on a convention that you deeply subscribe to. At this point, it does not seem like Chisholm has established that persons warrant special categorical distinction.

Bundle Theory and Ego Theory In the essay *Divided Minds and the Nature of Persons* [15], Parfit contrasts two competing views of persons, Bundle Theory and Ego Theory. Bundle Theory proposes that unity of consciousness cannot be explained by referring to persons; Bundle Theorists maintain that persons only exist as a feature of language. Persons are a convention—they are a *label* for a *bundle of things* like thoughts, memories, and sensations that are unified by causal kinds of relations. There are no subjects in which these bundles inhere. But the Ego Theorist proposes that these subjects really do exist—as subjects of experiences—and that they *have* a *bundle of things* like thoughts, memories, and sensations.

In support of Bundle Theory, Parfit presents the reader with a hypothetical situation in which a replica of you is created, atom by atom. At one end of the spectrum, if one-hundred percent of your atoms are replaced, this is clearly a *replica of you*. At the other end, if only one percent of your atoms are replaced, this is clearly *you*. Thus, in between these two extremes there must be a critical turning point where it goes from being *you* to being a *replica of you*. However, this appears implausible because any point chosen would seem arbitrary. How could a single atom make a difference? It is similar to the question, *When does sand become a pile of sand?* It is absurd to think that, at one point it is not a pile, and with the addition of a single grain of sand, it becomes a pile.

A practical illustration: The case of Phineas Gage Consider the famous case of Phineas Gage, a railroad construction foreman who survived an accident in 1848 where an iron rod passed through his skull, destroying much of his left frontal lobe. Before the accident, Gage was described as responsible, industrious, and well-liked. After recovering, he became impulsive, profane, and unable to hold steady employment. His friends said he was “no longer Gage.”

This case poses profound questions for both theories. For the Ego Theorist,

is the post-accident person the same *subject of experience* as before? If Gage’s personality, values, and behavioral patterns changed so dramatically, what persisted that makes him the same person? The Ego Theorist must claim there is some essential “Gage-ness” that survived despite these changes.

The Bundle Theorist has a different challenge: if Gage is just a bundle of psychological states and dispositions, and these changed dramatically, in what sense is he the same person? The Bundle Theorist might respond that there is sufficient psychological continuity—memories, basic cognitive abilities, learned skills—to maintain identity, but the degree of change raises questions about where to draw the line.

Ego Theorists have difficulty responding to this question because they believe that persons are *subjects of experiences*, not just *bundles of things*. As such, there must be some special unknown thing that makes you who you are that may not be included in the replication. Bundle Theorists have little problem answering these questions, however, because they deny that there is a *you*—only a *bundle of things*. If an exact copy of the *bundle of things* that constitutes you is made, what we choose to call this thing, *you* or a *replica of you*, is a matter of convention.

Split-brain studies and the nature of persons Parfit argues that empirical results from split-brain case studies provide evidence for Bundle Theory over Ego Theory. In pioneering research by Sperry [17] and Gazzaniga [9], patients who had their corpus callosum severed (to treat severe epilepsy) were presented with stimuli designed to reach only one hemisphere. When shown a placard whose left half is blue and right half is red, with each hemisphere receiving only one color, patients exhibited seemingly independent streams of consciousness—one aware of seeing only blue, the other aware of seeing only red.

Parfit takes such cases to pose a challenge for Ego Theory: if a person is essentially a subject of experience, and split-brain patients manifest two distinct subjects of experience, does this mean one person has become two? Or that there were always two persons sharing a body? Neither option seems palatable.

However, this argument overstates the difficulty for Ego Theory. Contemporary Ego Theorists have developed several responses that avoid Parfit’s dilemma:

The temporary fission response. Some Ego Theorists [14] argue that split-brain cases represent temporary, partial fission events. During tasks that elicit divided consciousness, there may indeed be two subjects of experience, but these are transient and incomplete. During normal activity, the hemispheres cooperate sufficiently that only one subject exists. The occurrence of temporary fission, while metaphysically interesting, does not undermine the Ego Theory’s central claim that persons are subjects of experience—it merely shows that in extreme circumstances, subjects can divide or merge.

The dominant hemisphere response. Another strategy maintains that one hemisphere (typically the left, language-dominant hemisphere) houses the person’s primary consciousness, while the right hemisphere’s responses represent sub-personal processing rather than a full subject of experience. On this view, the right hemisphere’s capacities, while sophisticated, do not constitute a complete person or subject. This response is supported by the fact that split-brain

patients typically report a unified sense of self after surgery, centered in the language-capable hemisphere.

The animalist response. Eric Olson [14] defends animalism—the view that we are fundamentally biological organisms, not subjects of experience or psychological bundles. On this view, the split-brain patient remains one organism throughout, even if consciousness is temporarily divided. Personal identity consists in biological continuity, not psychological unity. Split-brain cases may show that psychological unity can be disrupted without destroying the organism, and hence the person.

Bundle Theory does have resources to handle split-brain cases more straightforwardly: if persons are just bundles of experiences unified by causal relations, then split-brain cases simply involve bundles that are partially separated. There is no deep puzzle about whether one person becomes two, because persons were never fundamental entities in the first place—they are conventional designations for sufficiently unified bundles of experiences.

However, Bundle Theory faces its own difficulties with split-brain cases. If psychological unity is what constitutes personal identity, and split-brain patients show divided psychology, we need principled criteria for when a single person exists versus when there are two. How much psychological unity is sufficient? The Bundle Theorist’s answer risks being arbitrary or purely conventional.

The implications for moral responsibility These debates about personal identity and the nature of persons have profound implications for moral responsibility. If persons do not persist through time in any robust sense, or if they are merely conventional designations for bundles of experiences, then holding them responsible for past actions becomes problematic. How can we punish or praise a “person” for actions performed by what might be, metaphysically speaking, a different entity? The Ego Theorist’s subject of experience and the Bundle Theorist’s collection of mental states both face challenges in grounding the kind of persistent agency that moral responsibility seems to require.

3 Ascribing moral responsibility

“Responsibility: A detachable burden easily shifted to the shoulders of God, Fate, Fortune, Luck or ones neighbor. In the days of astrology it was customary to unload it upon a star.” —Ambrose Bierce, *The Devil’s Dictionary*, 1911

Under what circumstances can a moral agent’s conduct be ascribed moral responsibility? At a minimum, it would seem to require that its conduct be a product of being able to select, in a way that is unconstrained by causal agencies, a specific action among a set of possible actions—that is, a product of exercising a free will. For instance, an agent’s decision to go shopping could be a product of a free will if it was possible for the agent to have decided not to go shopping. Conversely, if an agent stumbles off a building and free falls towards the ground, the act of free falling is certainly not a product of free will. Under the circumstances, the agent had no choice but to fall.

Four-dimensionalism

It is often said that entities have location and extension in a three-dimensional space. That is, physical entities can be specified in a three-dimensional Cartesian coordinate system (see Figure 1), x , y , and z , which provides three spatial dimensions—respectively, length, height, and width, e.g., a cubic entity extends m units along the x , y , and z axes.

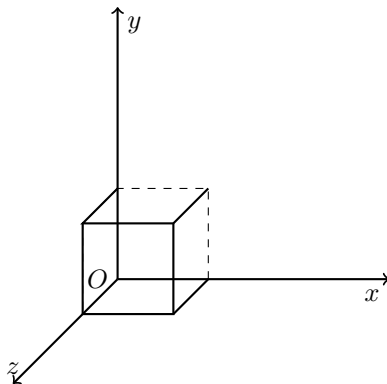


Figure 1: Three-dimensional Cartesian coordinate system with a unit cube illustrating spatial extension

However, our coordinate system is missing a crucial detail: at what time, t , will a particular entity be at a particular x , y , and z , and how long will it endure? Therefore, a physical entity's location and extension is more precisely specified in a four-dimensional coordinate system, x , y , z , and t , which provides three spatial dimensions and one temporal dimension, e.g., a hyper-cubic entity extends m units along the x , y , z , and t axes, or a person may extend an average of ≈ 0.6 meters along the x -axis, an average of ≈ 1.8 meters along the y -axis, an average of ≈ 0.3 meters along the z -axis, and ≈ 85.6 years along the t -axis.

Four-dimensional space-time has taken on a new-found importance with the advent of modern science. Under this model, when persons observe an entity at a particular time, they observe a single *time-slice* of it. More precisely, they observe a three-dimensional spatial “slice” (which is perpendicular to the temporal dimension) of a four-dimensional space-time entity. For example, when they observe a two-dimensional spatial slice of a sphere, they observe a disk; when they observe a three-dimensional spatial slice of a hyper-sphere, they observe a sphere.

At any time t , persons can only observe a three-dimensional spatial “slice” located at time t of a four-dimensional space-time entity. However, from a “God’s eye” vantage point outside the spacetime manifold, one could in principle observe the entire four-dimensional structure at once. From this perspective, the passage of time would represent different locations along the temporal dimension rather than genuine change.

Perdurantism and moral responsibility Does four-dimensionalism undermine moral agency? A common but mistaken objection holds that if persons are four-dimensional entities, they are “unchanging” and therefore cannot make genuine choices. This objection rests on a confusion. Four-dimensional entities do change—they have different temporal parts with different properties at different times. A person-worm extending from birth to death has temporal parts that differ in location, height, beliefs, desires, and moral character. The difference is that for the four-dimensionalist (or perdurantist), change consists in different temporal parts having different properties, rather than a single three-dimensional entity gaining and losing properties over time.

The more serious question is whether this view of persistence supports moral responsibility. Consider: if I am a four-dimensional worm extending through spacetime, and my temporal part at t_1 commits a crime, can my temporal part at t_2 be held responsible for it? The perdurantist answer is yes, provided the appropriate relations hold between these temporal parts—typically, psychological continuity and connectedness. My t_2 temporal part remembers the crime, has the same character that produced it, and is psychologically continuous with the t_1 part. This is sufficient for moral responsibility on most accounts.

Indeed, perdurantism may have advantages for explaining moral responsibility. It naturally accommodates cases of psychological change: if Phineas Gage’s pre-accident and post-accident temporal parts have very different psychological properties, we can say they are parts of the same four-dimensional person while acknowledging their profound differences. The question becomes whether there is sufficient continuity between temporal parts to ground responsibility, not whether a persisting three-dimensional entity has changed too much.

Eternalism, foreknowledge, and choice A distinct but related issue concerns the relationship between four-dimensionalism and the nature of time. Many four-dimensionalists accept eternalism (the B-theory of time)—the view that past, present, and future events are equally real, and temporal passage is illusory. On this view, facts about what you will choose tomorrow already exist, though they are located at a different temporal location than the present.

Does this undermine free choice? If it is already a fact that you will choose coffee tomorrow, could you have chosen tea? This raises the ancient problem of foreknowledge and freedom. Some argue that if future facts exist timelessly, choice is illusory—you cannot make it the case that you choose coffee, because this is already a fact written into the four-dimensional spacetime structure.

Others respond that eternalism is perfectly compatible with free choice. The fact that your choice already exists in the four-dimensional block does not mean it is determined by prior events. The eternalist can distinguish between: (1) it being fixed which choice you will make, and (2) your choice being causally determined. Even if future facts exist, they might exist precisely because you will freely choose them, not because they are forced by prior causes. The future choice is fixed in the sense that it will happen, but this is compatible with its being free in the sense that it flows from your autonomous deliberation rather than external compulsion.

As Sider [16] argues, four-dimensionalism is a thesis about the persistence

of objects through time, while eternalism is a thesis about the reality status of times themselves. Neither directly entails determinism. A four-dimensional person-worm can make genuinely free choices at different temporal locations, and these choices can be real facts without being causally determined facts.

Causal Determinism

Supposing that reality is not an immutable four-dimensional space-time, let us turn our attention to the question of causal determination. Causal determinism is the doctrine that all events are the inevitable result of antecedent conditions, including the actions of *persons*. The totality of existence is a game of cosmic billiards where, given an initial state as input, the final state could be output by a computable function.

The Laplacian demon Pierre-Simon Laplace [12] famously articulated the determinist position through his thought experiment of an omniscient intelligence—now known as Laplace’s demon. This hypothetical being, knowing the precise location and momentum of every atom in the universe at a given moment, could use the laws of physics to reconstruct the entire past and predict the entire future. In Laplace’s words, “nothing would be uncertain and the future, as the past, would be present to its eyes.”

This mechanistic view gained considerable support from classical physics, where Newton’s laws seemed to govern all motion with mathematical precision. Every effect had a cause, every cause had a prior cause, stretching back in an unbroken chain to the beginning of time. Human actions, being physical events in a physical brain, would be no exception to this universal causation.

Is causal determinism compatible with moral agency? In causal determinism, there is only one possible future; acts of apparent choice are the mechanical expression of a supposed moral agent’s heredity and past environment. Since its future is determined, that would denote that moral responsibility is not possible, for at the very least a moral agent must have the power to make choices between morally *good* and *bad* actions.

The incompatibilist position, defended by philosophers like van Inwagen [20], holds that determinism and free will cannot coexist. If our actions are the inevitable result of the laws of nature and events in the remote past—events over which we had no control—then we cannot be morally responsible for them. This is formalized in van Inwagen’s Consequence Argument: no one has power over facts about the distant past; no one has power over the laws of nature; therefore, no one has power over facts about the future (including their own actions).

The compatibilist alternative Not all philosophers accept the incompatibilist conclusion that determinism undermines moral responsibility. Indeed, compatibilism has historically been the majority position in philosophy and remains influential in contemporary debates. Compatibilists argue that free will and moral responsibility are compatible with causal determinism, provided we properly understand what freedom and responsibility require.

Classical compatibilism: Hume’s analysis. The classical compatibilist position, articulated by David Hume [10], argues that freedom is not the absence of

causation but rather a particular kind of causation. An action is free when it flows from the agent's own desires, beliefs, and character, rather than from external compulsion or constraint. On this view, "liberty" requires "necessity"—that is, for an action to be attributable to an agent, it must necessarily follow from that agent's psychological states.

Consider the difference between two scenarios: (1) You decide to donate to charity because you value helping others and believe this charity does important work; (2) You donate to charity because someone holds a gun to your head and threatens to shoot if you don't. In the first case, your action flows from your own values and reasoning—this is paradigmatically free, even though your values and reasoning were themselves shaped by prior causes (your upbringing, experiences, etc.). In the second case, your action is compelled by external threat—this is unfree. The compatibilist claims this distinction is all the freedom worth wanting, and determinism does not collapse it.

Frankfurt's hierarchical model. Harry Frankfurt [7, 8] developed a sophisticated compatibilist account centered on second-order desires. On his view, persons are distinguished from other agents by their capacity to form desires about their desires—to want to want certain things. Freedom consists in having one's will (first-order desires that move one to action) aligned with one's second-order desires about what one's will should be.

Frankfurt's famous thought experiments challenge the principle of alternate possibilities—the idea that moral responsibility requires the ability to do otherwise. Consider a case where a neurosurgeon has implanted a device in your brain that will make you choose coffee if you start to choose tea, but the device never activates because you choose coffee on your own. Frankfurt argues you are morally responsible for choosing coffee even though you could not have done otherwise, because the choice flowed from your own reasoning and values. What matters is not whether alternative possibilities exist, but whether you act on your own authentic reasons.

Reasons-responsiveness theories. John Martin Fischer and Mark Ravizza [6] developed an influential compatibilist account based on reasons-responsiveness. On their view, an agent is morally responsible for an action if the mechanism that produces the action is responsive to reasons—that is, if the agent would have acted differently if presented with sufficient reason to do so. This approach allows for moral responsibility even in a determined world, as long as the causal processes governing behavior are sensitive to rational considerations.

For instance, suppose determinism is true and your decision to help a drowning child was causally necessitated by prior events. Fischer and Ravizza argue you are still morally responsible if the mechanism producing your action (your practical reasoning) is such that, had there been overwhelming reasons not to help (e.g., helping would cause ten other children to drown), you would have responded to those reasons and acted differently. The actual causal necessitation does not undermine responsibility as long as the mechanism is reasons-responsive.

The randomness objection answered. Compatibilists also argue that indeterminism provides no advantage for moral responsibility. If your decision to

help or harm someone depends on genuinely random quantum events in your neurons, how does this make you more responsible than if the decision depends on your character and values? Randomness seems to undermine responsibility rather than enable it. As Daniel Dennett [5] argues, the alternative to determinism is not freedom but chaos. The libertarian’s indeterministic agent seems less like a responsible person and more like a roulette wheel.

Challenges to compatibilism. Despite its sophistication, compatibilism faces significant objections. Incompatibilists argue that even if an action flows from one’s character, if that character was itself determined by factors outside one’s control (genetics, early environment, etc.), then the agent is not truly responsible. Galen Strawson [18] presents what he calls the “Basic Argument”: to be truly responsible for an action, you must be responsible for the way you are (your character, values, etc.); but to be responsible for the way you are, you must have chosen to be that way; but to choose to be a certain way, you must already have a character that makes that choice; this leads to infinite regress or an arbitrary starting point for which you are not responsible.

Moreover, some argue that compatibilism changes the subject—that when ordinary people think about free will and moral responsibility, they have in mind a libertarian conception incompatible with determinism, and compatibilists offer a different, weaker concept and simply label it “freedom.” Whether this objection succeeds depends partly on empirical questions about folk concepts and partly on normative questions about what freedom is worth wanting.

Quantum indeterminacy and libertarian free will The advent of quantum mechanics has introduced fundamental indeterminacy into our picture of reality. At the quantum level, events may be genuinely probabilistic rather than determined. Some libertarians about free will have seized on this as providing the metaphysical room needed for genuine choice. However, critics point out that random quantum events hardly seem a better foundation for moral responsibility than deterministic ones. If your decision to help or harm someone depends on quantum coin-flips in your neurons, how does this make you more responsible than if it depends on prior causes?

Robert Kane [11] has developed a sophisticated libertarian account that attempts to navigate between determinism and randomness. On his view, quantum indeterminacy may play a role in moments of moral conflict, where we experience genuine uncertainty about what to do. The resolution of this conflict, while not determined, is not merely random but shaped by our efforts of will. The indeterminacy provides the metaphysical space for self-formation, where we become responsible for our character through these undetermined but non-random choices.

An argument from absurdity: the coherency of choice For the sake of argument, assume choice is not an illusion. How then might we offer a rational account of it? But if something is rational, does that not signify that it can be explicated in terms of its antecedent conditions? How else can a physical process be rationally explained if not by cause-and-effect? But if choice can be understood in terms of its antecedent conditions, then it is not free—it was determined by cause-and-effect.

This dilemma—that free will seems impossible whether determinism is true or false—has led some philosophers to hard incompatibilism. Derk Pereboom and others argue that free will is incompatible both with determinism and with indeterminism, and therefore moral responsibility in the traditional sense is impossible. Yet even these skeptics often argue that we can maintain many moral practices and attitudes without ultimate moral responsibility. We can still shape behavior through praise and blame, protect society from dangerous individuals, and express moral emotions like guilt and indignation—understanding these as forward-looking practices rather than backward-looking judgments of ultimate desert.

4 Concluding thoughts

The metaphysical challenges to moral responsibility examined in this paper reveal deep tensions in our understanding of persons as moral agents. We have seen how the very foundations of moral agency—the existence of persons as persistent subjects, the reality of choice, and the possibility of acting otherwise—face serious philosophical challenges.

The cumulative challenge The metaphysical challenges examined in this paper operate on multiple levels, and their interaction creates compounding difficulties for moral responsibility. We can distinguish three levels of challenge:

Level 1: The reality of moral properties. If Bradley’s regress is vicious and realism about Universals fails, then moral properties lack objective ontological status. Even if nominalism provides an adequate account of descriptive properties, the extension to normative properties remains questionable. If “moral agent” is a conventional classification rather than a natural kind, the metaphysical foundation for moral responsibility appears threatened.

Level 2: The persistence of moral agents. Even if we grant that moral properties exist, we must ask: what entities bear these properties? If Bundle Theory is correct and persons are conventional designations for causally related bundles of experiences, then personal identity across time becomes conventional. But moral responsibility seems to require that the person punished today is genuinely the same person who committed the crime yesterday, not merely conventionally the same. Split-brain studies and cases like Phineas Gage raise questions about how much psychological change is compatible with continued identity. If personal identity admits of degrees or borderline cases, does moral responsibility similarly admit of degrees?

Level 3: The freedom to choose. Even granting objective moral properties and persisting persons, we must ask whether persons can freely choose their actions. If causal determinism is true and incompatibilism is correct, then no one is morally responsible for anything. Four-dimensionalism and eternalism raise questions about whether choice is metaphysically possible if future facts already exist. Libertarian indeterminism may provide metaphysical space for choice, but faces the challenge that random events ground responsibility no better than determined ones.

These challenges interact in complex ways. Consider the implications of different combinations of views:

- If realism + Ego Theory + libertarianism are all true, moral responsibility has maximum metaphysical support: objective moral properties, persistent subjects, and genuine choice.
- If nominalism + Bundle Theory + compatibilism are all true, moral responsibility becomes thoroughly naturalized: conventional moral categories, conventional persons, and freedom understood as reasons-responsiveness. This may be sufficient for moral practice but lacks the robust metaphysical foundation many assume.
- If nominalism + Bundle Theory + hard incompatibilism are all true, traditional moral responsibility collapses entirely, though forward-looking practices of praise, blame, and punishment might be retained on pragmatic grounds.

Importantly, one can accept some challenges while resisting others. A philosopher might accept nominalism about moral properties while defending Ego Theory and libertarianism, thereby salvaging much of moral responsibility despite losing objective moral properties. Or one might accept Bundle Theory and determinism while defending compatibilism, preserving moral responsibility despite conventionalist personal identity and causal necessity.

Pragmatic implications Yet the practical question remains: what would be entailed by widespread acceptance that moral agency is not a legitimate concept? Historical evidence suggests that societies do not simply collapse when confronted with challenges to their fundamental assumptions. The scientific revolution removed Earth from the center of the universe without ending human civilization. Darwin's theory of evolution challenged human exceptionalism without eliminating ethics.

Similarly, the metaphysical challenges to moral responsibility need not lead to nihilism or social breakdown. Even hard incompatibilists like Pereboom argue for maintaining many moral practices on pragmatic grounds. We might reconceive punishment as rehabilitation and deterrence rather than retribution. We might understand moral emotions as evolutionary adaptations that promote cooperation rather than responses to ultimate desert. We might treat character development as a causal process subject to influence rather than a product of libertarian free will.

The persistence of moral phenomenology Perhaps most significantly, the phenomenology of moral agency—the felt experience of deliberation, choice, and responsibility—persists regardless of metaphysical arguments. Even those intellectually convinced that free will is an illusion continue to experience decision-making as involving genuine alternatives. This experiential reality may be more fundamental to moral practice than metaphysical truth.

P.F. Strawson [19] argued that our practices of holding people responsible are grounded not in metaphysical facts but in reactive attitudes like resentment,

gratitude, and indignation. These attitudes are constitutive of moral responsibility rather than responses to it. On this view, the question is not whether people *are* moral agents in some ultimate metaphysical sense, but whether treating them as such is integral to the forms of life we value.

Assessment of the challenges Where does this investigation leave us? I have argued that moral responsibility faces significant metaphysical challenges on three fronts, though these challenges differ in severity:

The challenge from properties. The question of whether moral properties are objective Universals or conventional classifications remains genuinely unresolved. Bradley's regress poses real difficulties for realism, but the realist responses—particularly Armstrong's non-relational tie strategy—are not obviously unsuccessful. Nominalism avoids Bradley's regress but may struggle to capture the normativity of moral properties. This challenge is serious but not fatal; both realists and nominalists can construct coherent accounts of moral properties, though neither is entirely satisfactory.

The challenge from personal identity. The debate between Bundle Theory and Ego Theory also remains open, with sophisticated defenses of both positions. Split-brain studies and cases of radical psychological change raise genuine puzzles, but neither view is clearly defeated by the empirical evidence. Importantly, both views can ground sufficient continuity for moral responsibility in typical cases. The challenge here is more about borderline cases and theoretical elegance than about whether moral responsibility is possible at all.

The challenge from freedom. This is the most serious metaphysical challenge to moral responsibility. If hard incompatibilism is correct—if free will is incompatible with both determinism and indeterminism—then moral responsibility in the traditional sense is impossible. However, compatibilism remains a viable and well-developed position. If compatibilists are correct that responsibility requires only reasons-responsiveness and acting from one's character rather than libertarian freedom, then the challenge from determinism dissolves. The debate between compatibilists and incompatibilists is unlikely to be resolved definitively, but this means the case against moral responsibility from determinism is not conclusive.

The way forward Several philosophical paths remain open for defending moral responsibility despite these challenges:

The naturalized approach (nominalism + Bundle Theory + compatibilism): Accept that moral properties, personal identity, and freedom are all naturalistic phenomena that may be partially conventional but are nonetheless real and important. On this view, moral responsibility does not require robust metaphysical foundations—it requires only that persons (conventionally understood) can respond to reasons and that holding them responsible serves important social functions.

The partially grounded approach: Accept realism about some elements while embracing conventionalism about others. For instance, one might be a realist about personal identity (Ego Theory) and a compatibilist about freedom, while accepting that moral categories are partially conventional. Or one might be a moral realist but accept Bundle Theory, arguing that bundles can bear objective

moral properties.

The pragmatic approach (Strawson): Following P.F. Strawson [19], argue that moral responsibility is grounded not in metaphysical facts but in reactive attitudes—resentment, gratitude, indignation—that are constitutive of moral practice. On this view, the metaphysical questions examined in this paper, while intellectually interesting, are not directly relevant to whether moral responsibility exists. What matters is whether treating people as responsible agents is integral to the forms of life we value.

Final reflection This investigation reveals that moral responsibility faces genuine metaphysical challenges, but these challenges are not insurmountable. The strongest position may be a modest one: moral responsibility exists, but its metaphysical foundations are more complex, more naturalistic, and more dependent on convention than many pre-theoretically assume. We need not choose between full-throated metaphysical realism about moral agency and complete eliminativism. A middle path—acknowledging both the conventional aspects of moral categories and their genuine importance in human life—may be the most defensible position.

The ancient question “Are persons moral agents?” admits of no simple answer. The answer depends on what we mean by “persons,” what we mean by “moral,” and what metaphysical commitments we are willing to accept. But this complexity need not lead to skepticism. Understanding the metaphysical underpinnings of moral responsibility—even if they turn out to be partially conventional, naturalistic, and pragmatic—may enhance rather than undermine our moral practices, by helping us understand what we are really doing when we hold each other responsible.

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